RAJALAKSHMI ENGINEERING COLLEGE

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2027 IEEE Madhya Pradesh Section Conference : Submission (156) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org> Tue, Oct 13, 2026 at 12:10 AM
Reply-To: Microsoft CMT - Do Not Reply <noreply@msr-cmt.org>
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Hello,

The following submission has been edited.

Track Name: Track 1: Artificial Intelligence and Machine Learning

Paper ID: 156

Paper Title: Haya: Live-Compute-Index-Guided, Swap-Cost-Aware Routing for On-Device

Abstract:
The primary trade-off involves the level of task-device
state alignment and the associated overhead for resident model
update. This paper describes Haya, an offline solution that: (i)
calculates the Live Compute Index (LCI) for each candidate; (ii)
performs feasibility tests with respect to the available memory;
and (iii) finds out the amortized overhead required to switch
models and reconstruct history for each candidate, making a
switch possible only if the gains achieved in both cases exceed the hysteresis threshold. Three Q4_0
models have been considered by using a replay dataset containing 200 prompts on Realme GT 6T and Realme 11
Pro+ devices. Compared to Complexity-Only,
Haya decreases latency and energy consumption by 29.6% and
17.2%, respectively, on GT 6T, and 32.1% and 20.7%,
respectively, on Realme 11 Pro+, decreasing the number of
switches from 41 to 11, and from 44 to 13, respectively. Humanjudgment quality in terms of means is 0.746
as compared to 0.724 on GT 6T, and 0.718 as compared to 0.724 on Realme 11 Pro+. An experiment conducted
on GT 6T shows that without considering the swap-cost factor, the number of model switches increases from
11 to 38, and the average latency increases from 3,253 ms to 3,820ms. Timings show that reconstruction of
1,024 history tokens requires 85% of the total time 29.12 seconds allocated for handoff.
Index Terms—On-device inference, model routing, model
switching, swap cost, smartphone language models.

Created on: Mon, 12 Oct 2026 18:39:09 GMT

Last Modified: Mon, 12 Oct 2026 18:40:48 GMT

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Primary Subject Area: LLM and AI

Secondary Subject Areas:
Edge-Computing

Submission Files:
Haya_final_paper_LCI_IEEE (3).pdf (744 Kb, Mon, 12 Oct 2026 18:37:50 GMT)

Submission Questions Response: Not Entered

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